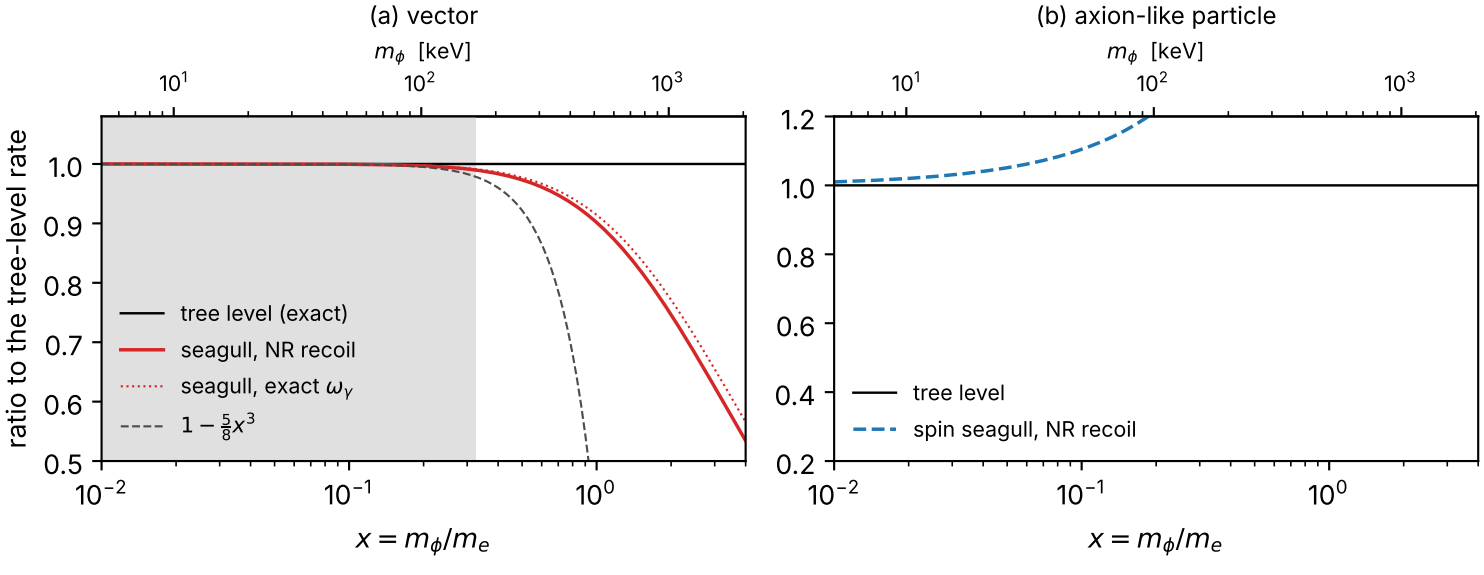
(a) vector
m_ϕ [keV]
10^1
10^2
10^3
ratio to the tree-level rate
1.0
0.9
0.8
0.7
0.6
0.5
tree level (exact)
seagull, NR recoil
seagull, exact ω_γ
$1-\frac{5}{8}x^3$
10^{-2}
10^{-1}
10^0
$x = m_\phi/m_e$
(b) axion-like particle
m_ϕ [keV]
10^1
10^2
10^3
1.2
1.0
0.8
0.6
0.4
0.2
tree level
spin seagull, NR recoil
10^{-2}
10^{-1}
10^0
$x = m_\phi/m_e$